Aggregate Functions

```
SQL*Plus: Release 11.2.0.1.0 Production on Tue Aug 11 02:31:35 2026

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Enter user-name: A020@db2021
Enter password:

Connected to:
Oracle Database 11g Release 11.2.0.1.0 - Production

SQL> CREATE TABLE STUDENT55(
  2  STUDENT_ID NUMBER(5) PRIMARY KEY,
  3   STUDENT_NAME VARCHAR2(30),
  4   DEPARTMENT VARCHAR2(20),
  5  MARKS NUMBER(3)
  6  );

Table created.
SQL> insert into student55 values (101, 'Gopi','cse',85);

1 row created.

SQL> insert into student55 values (102, 'kiran','ece',78);

1 row created.

SQL> insert into student55 values (103, 'ria','it',92);

1 row created.

SQL> insert into student55 values (104, 'nandhini','cse',88);

1 row created.

SQL> insert into student55 values (105, 'swathi','ece',81);

1 row created.

SQL> insert into student55 values (106, 'tulsi','it',95);

1 row created.

SQL> select count(*) as total_students
  2  from student55;

TOTAL_STUDENTS
--------------
             6

SQL> select sum(marks) as total_marks from student55;

TOTAL_MARKS
-----------
        519

SQL> select avg(marks) as average_marks from student55;

AVERAGE_MARKS
-------------
         86.5

SQL> select max(marks) as highest_mark from student55;

HIGHEST_MARK
------------
          95

SQL> select min(marks) as lowest_mark from student55;

LOWEST_MARK
-----------
         78

SQL> select department, avg(marks) as avg_marks from student55 group by department;
DEPARTMENT            AVG_MARKS
```

```
-------------------- ----------
cse                        86.5
it                         93.5
ece                        79.5

SQL> select department, count(*) as total_students from student55 group by deparment;
select department, count(*) as total_students from student55 group by deparment
                                                                      *
ERROR at line 1:
ORA-00904: "DEPARMENT": invalid identifier


SQL> select department, count(*) as total_students from student55 group by department;

DEPARTMENT           TOTAL_STUDENTS
-------------------- --------------
cse                               2
it                                2
ece                               2

SQL> select department, max(marks) as max_marks from student55 group by department;

DEPARTMENT            MAX_MARKS
-------------------- ----------
cse                          88
it                           95
ece                          81

SQL> select department, count(*) as total_students from student55 group by department having count(*) > 1;

DEPARTMENT           TOTAL_STUDENTS
-------------------- --------------
cse                               2
it                                2
ece                               2

SQL> SELECT DEPARTMENT, AVG(MARKS) AS AVG_MARKS
  2  FROM STUDENT
  3  GROUP BY DEPARTMENT
  4  HAVING AVG(MARKS) > 85;
HAVING AVG(MARKS) > 85
           *
ERROR at line 4:
ORA-00904: "MARKS": invalid identifier


SQL> select department, avg(marks) as avg_marks from student55 group by department having avg(marks) > 85;

DEPARTMENT            AVG_MARKS
-------------------- ----------
cse                        86.5
it                         93.5

SQL> select department, count(*) as no_ of_students, avg(marks) as avg_marks, max(marks) as highest_mark f
    rom student55 group by department having avg(marks) > 80;
select department, count(*) as no_ of_students, avg(marks) as avg_marks, max(marks) as highest_mark from s
    tudent55 group by department having avg(marks) > 80
                                       *
ERROR at line 1:
ORA-00923: FROM keyword not found where expected


SQL> select department, count(*) as no_of_students, avg(marks) as avg_marks, max(marks) as highest_mark fr
    om student55 group by department having avg(marks) > 80;

DEPARTMENT           NO_OF_STUDENTS  AVG_MARKS HIGHEST_MARK
-------------------- -------------- ---------- ------------
cse                               2       86.5           88
it                                2       93.5           95

SQL> select * from student55;

STUDENT_ID STUDENT_NAME                   DEPARTMENT                MARKS
---------- ------------------------------ -------------------- ----------
       101 Gopi                           cse                           85
       102 kiran                          ece                           78
       103 ria                            it                            92
```

```
       104 nandhini                  cse                       88
       105 swathi                    ece                       81
       106 tulsi                     it                        95

6 rows selected.
```